

CMOS Logic Circuits and Family

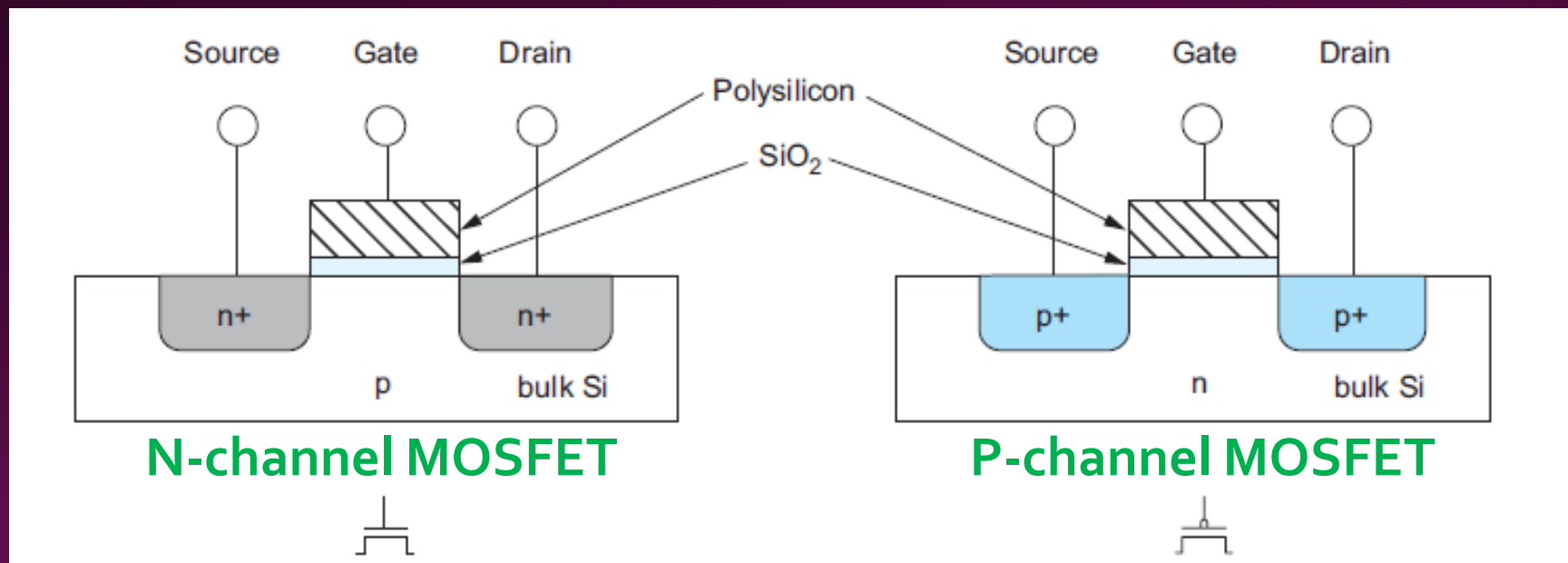
By Prof. Tanmoy Biswas,
Department of computer science,
Syamaprasad college.

Objective

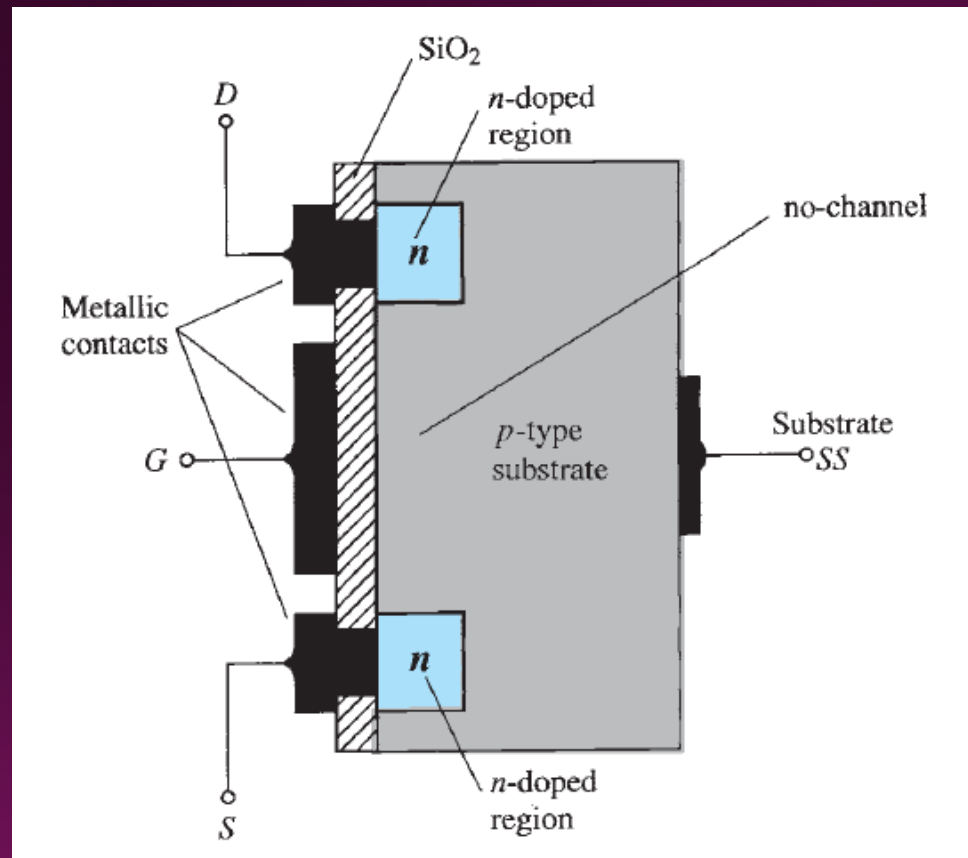
- MOS Technology
- Structure, construction, threshold voltage and concept of inversion layer in enhancement type MOSFET.
- MOS Inverters – Active and passive load
- CMOS inverters, NAND and NOR gates.
- CMOS characteristics.
- CMOS – TTL compatibility.

MOS Transistors

Enhancement type MOSFET are basically used to realize digital logic and used in construction of logical circuits.

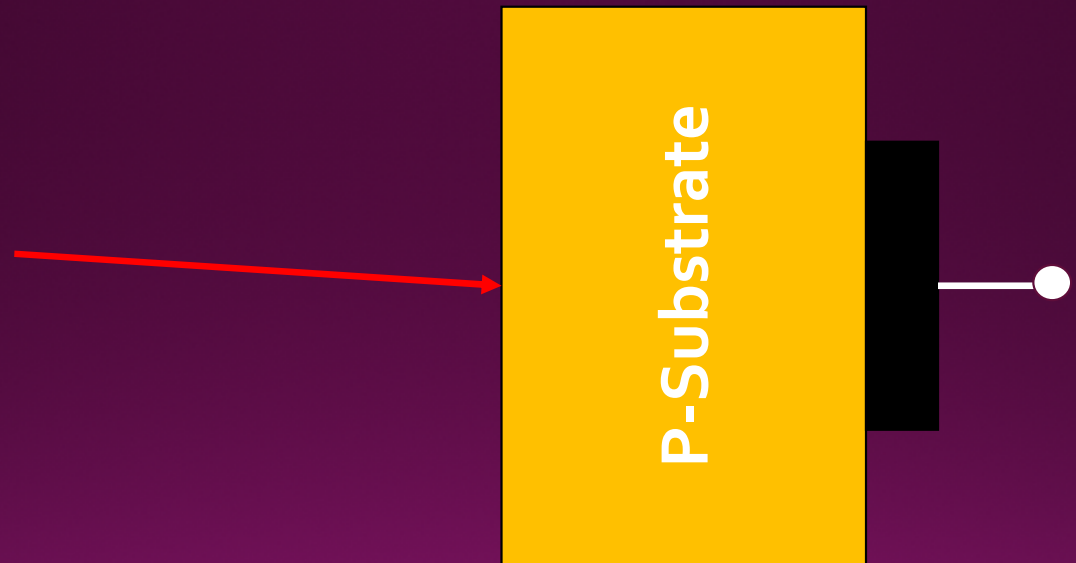


MOS Transistors – Overview of Construction



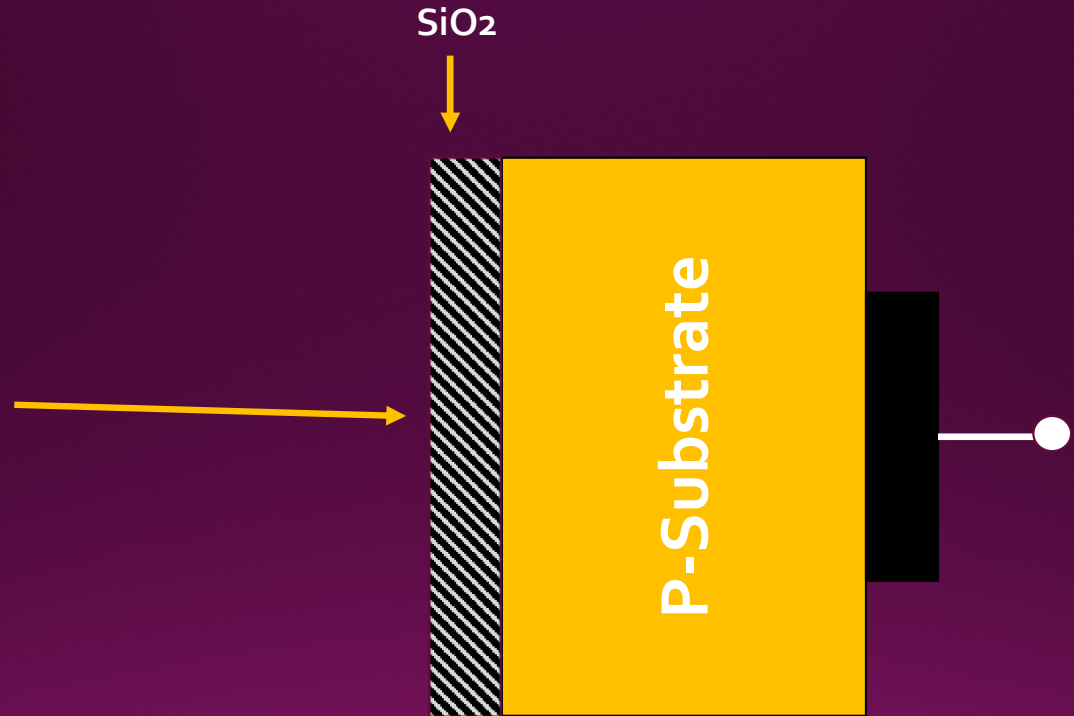
MOS Transistors - Construction

Step1: p-type substrate is taken if **n-channel** needs to be formed.



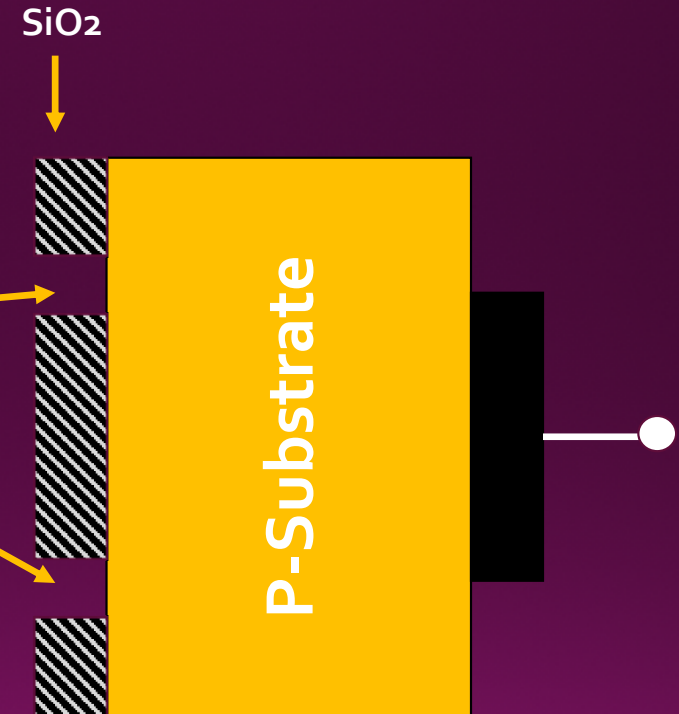
MOS Transistors - Construction

Step 2: A layer of SiO_2 layer is formed over the p substrate.



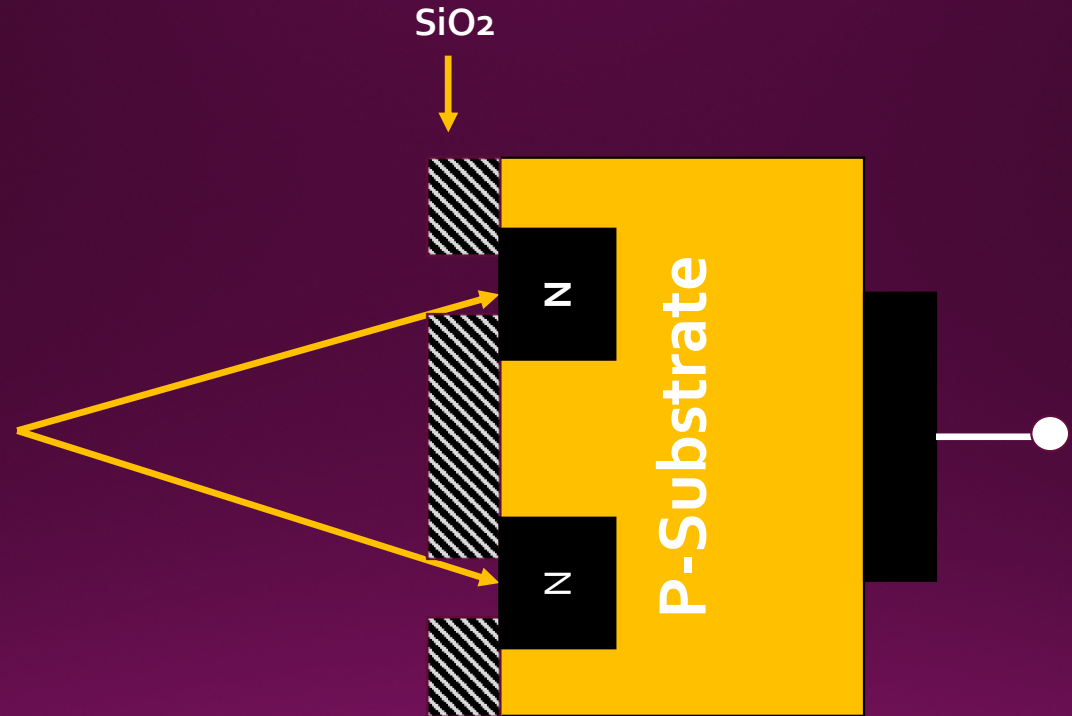
MOS Transistors - Construction

Step 3: Etching of layer of SiO_2 is done.



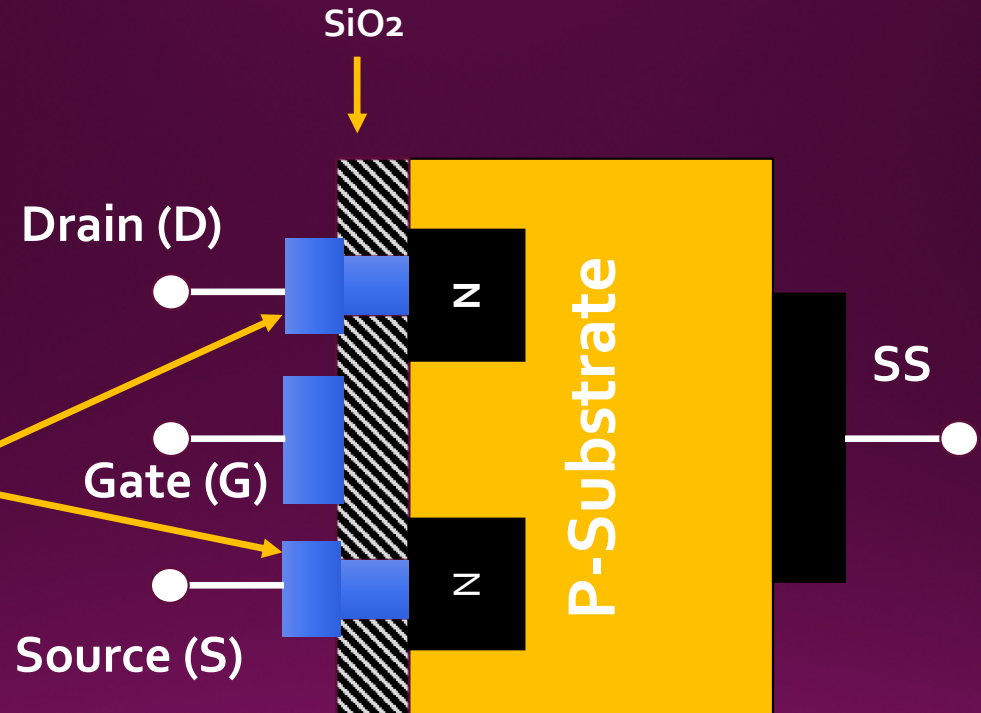
MOS Transistors - Construction

Step 4: Doping of N-Well



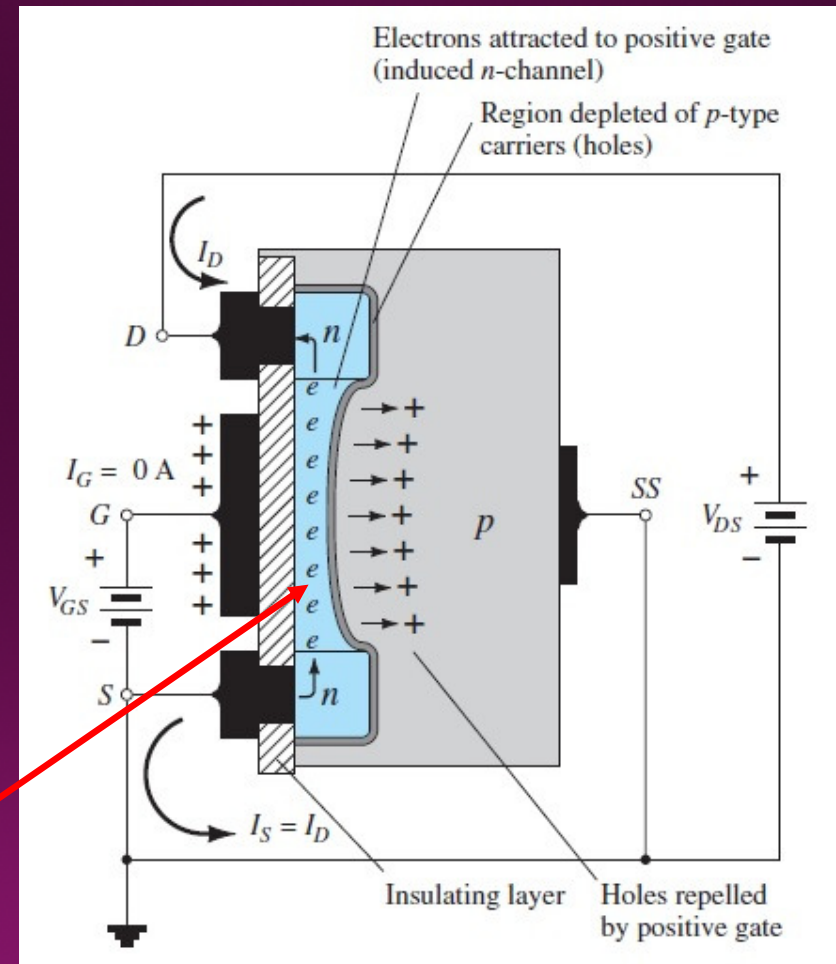
MOS Transistors - Construction

Step 5: Metallization is carried out for terminal connectivity.

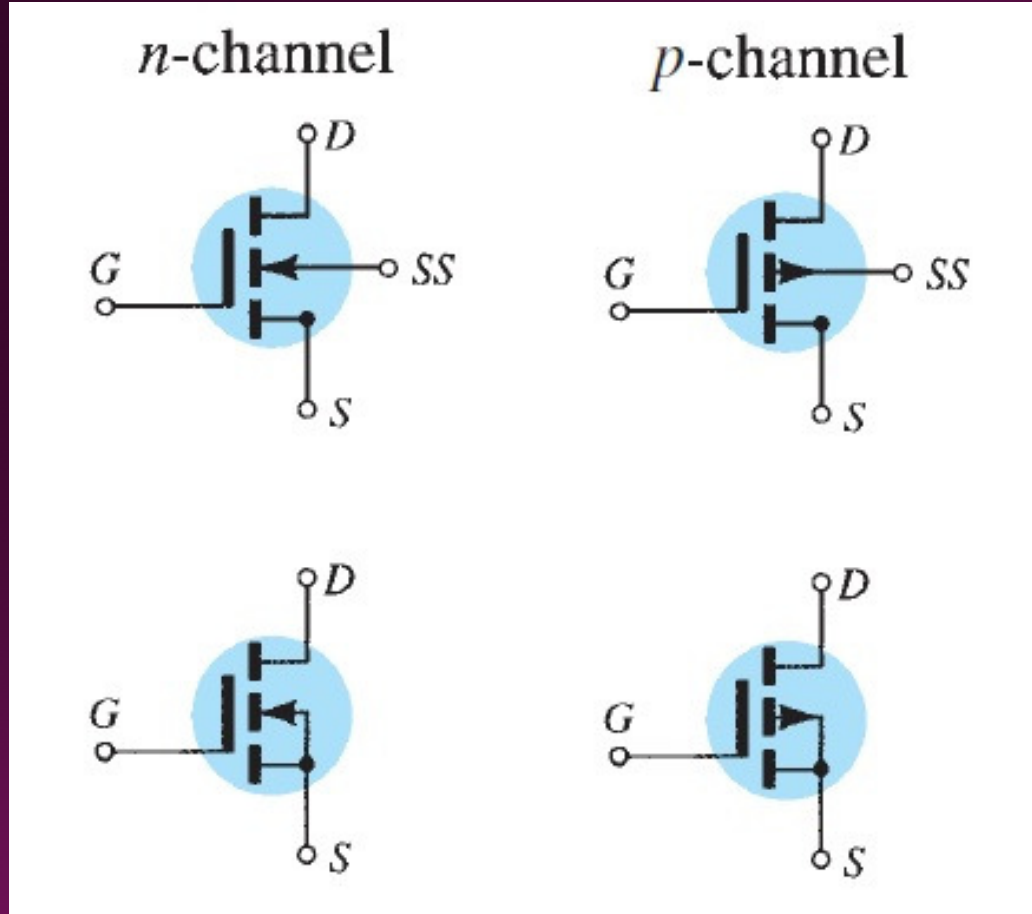


MOS Transistors – Working Principle

- When a V_{GS} is connected and the MOSFET is normally biased (V_{DS}) positive layer over the dielectric layer pushes the holes and attracts the electrons.
- The Gate to Source (V_{GS}) voltage at which the channel is formed between drain and source is called Threshold voltage (V_{TH}).
- The layer created is called an **inversion layer**.

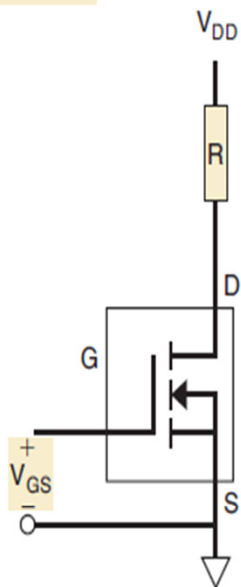


MOS Transistors – Symbol

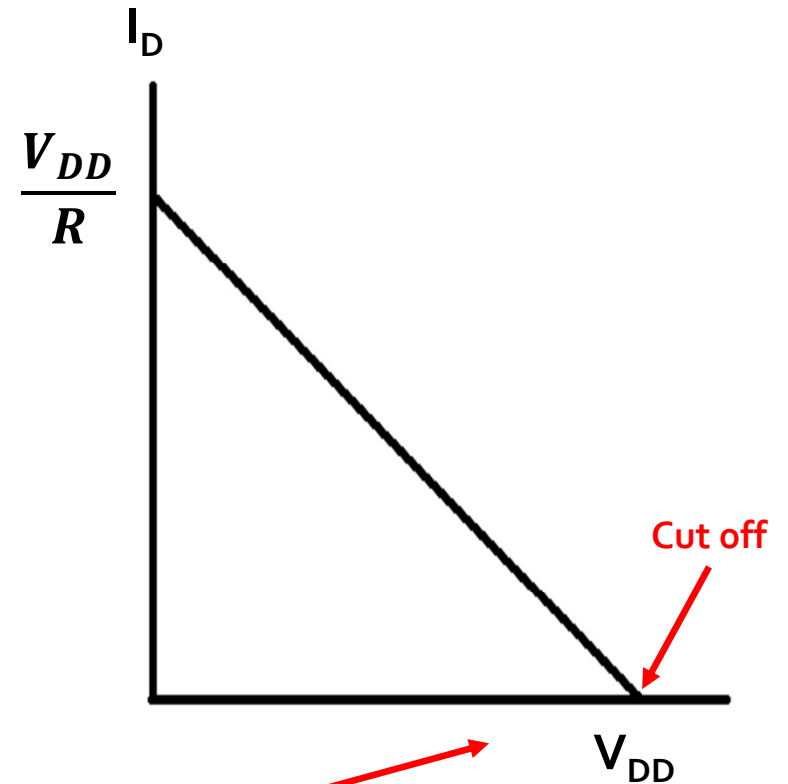
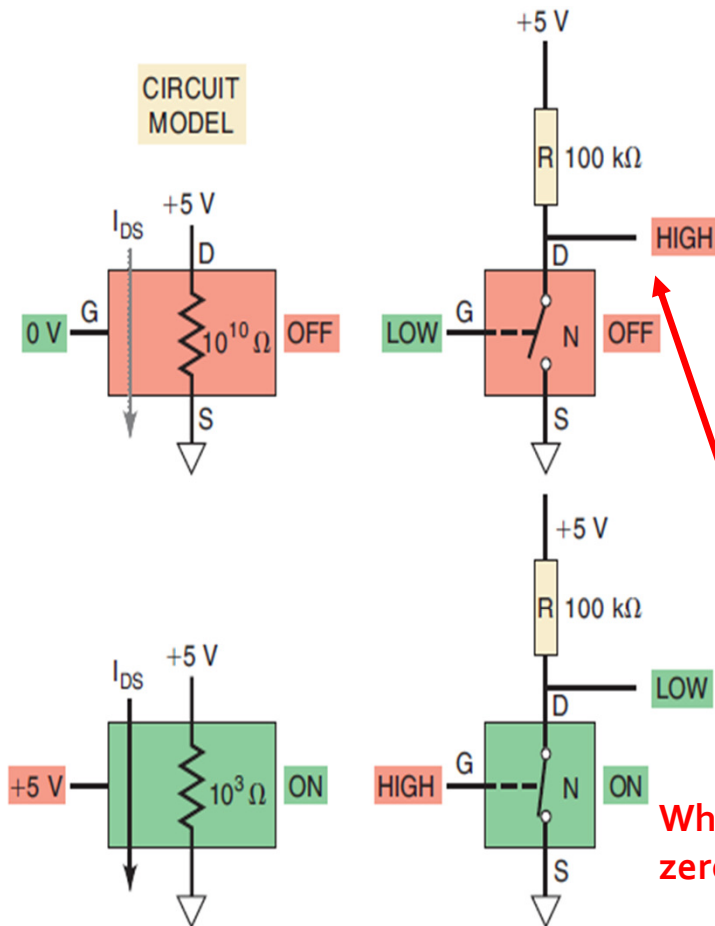


MOS Inverter – Passive load

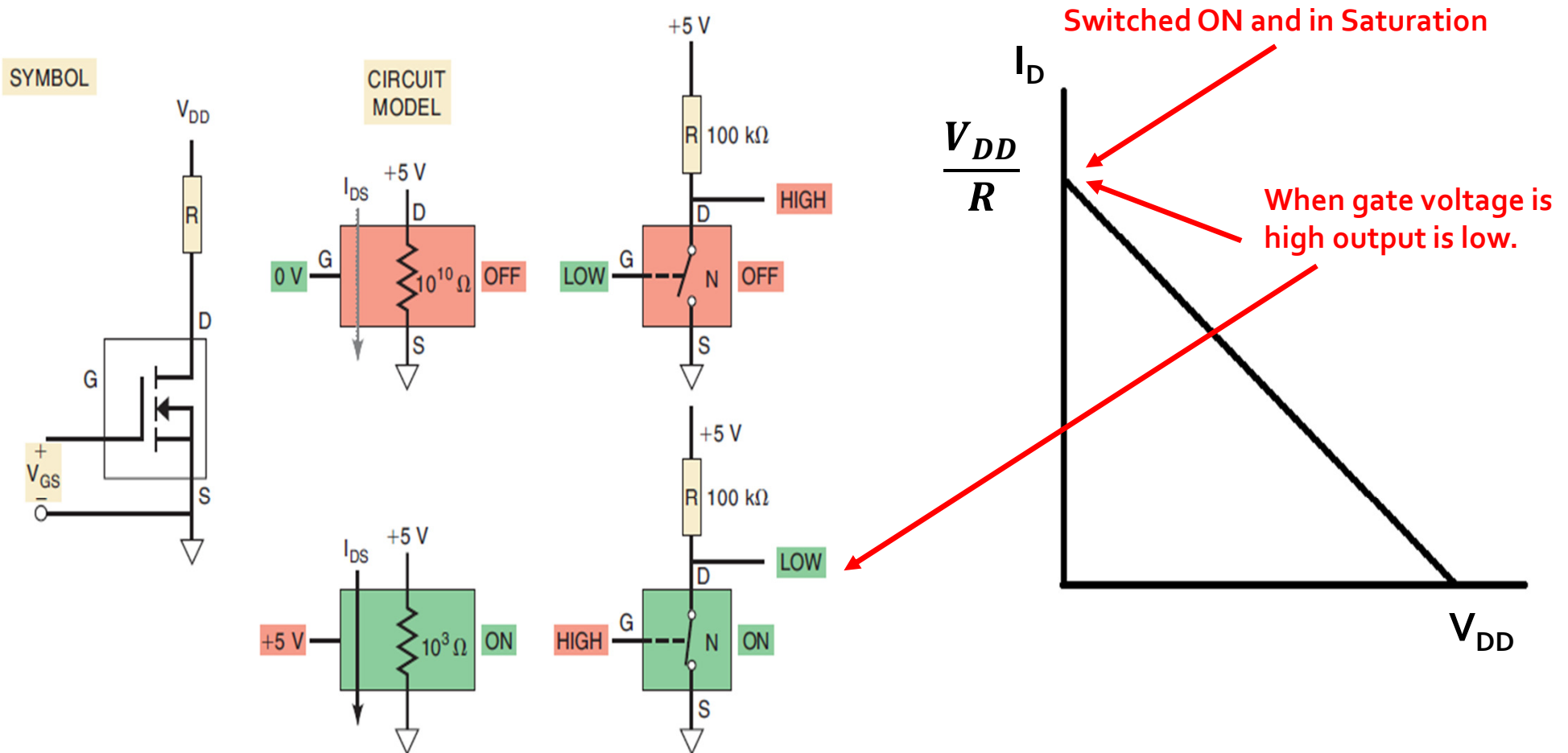
SYMBOL



CIRCUIT MODEL

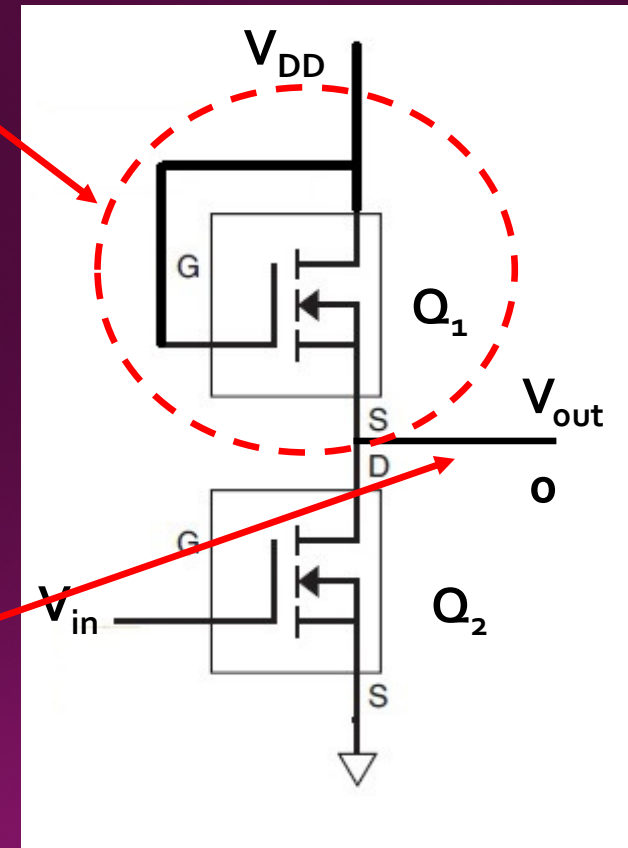


MOS Inverter – Passive load



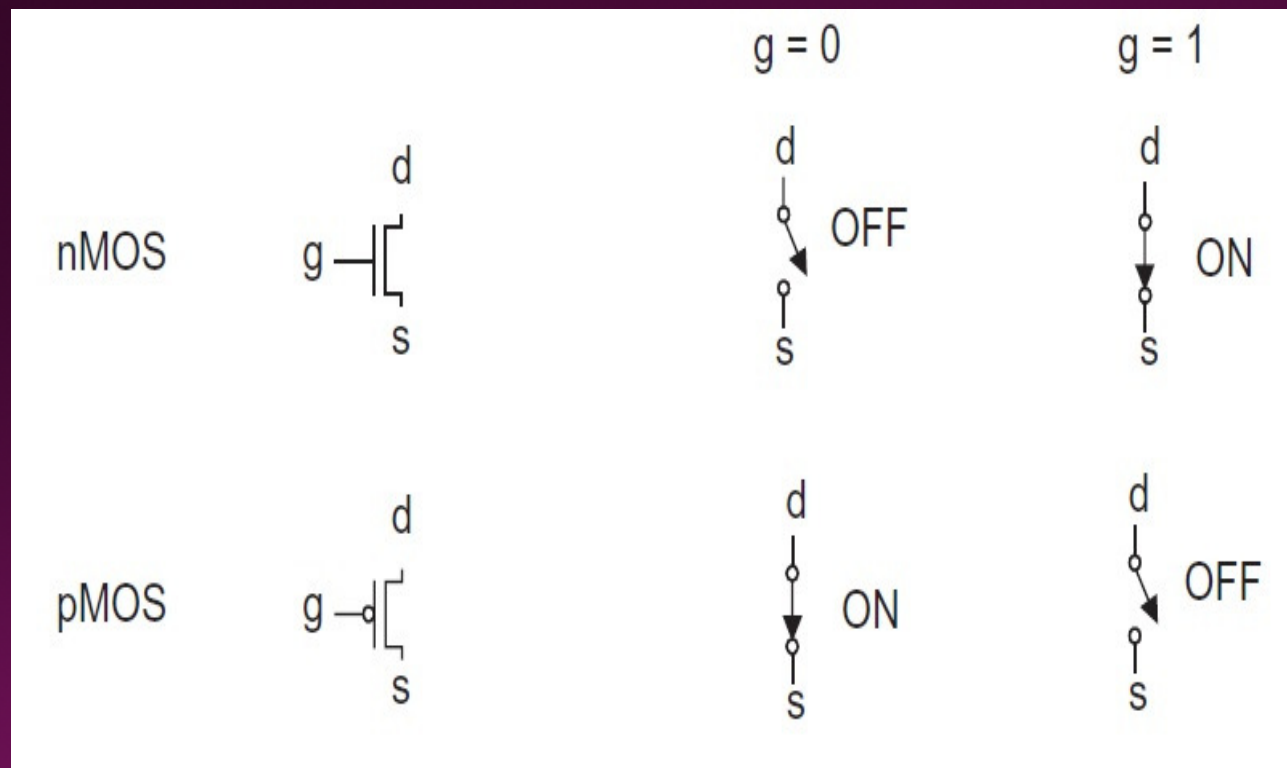
MOS Inverter – Active load

- The MOSFET Q_1 acts as a load resistance.
- The gate is tied to the supply voltage which actually helps in creating the channel inside the MOSFET.
- When an input voltage is provided at V_{in} the lower MOSFET Q_2 conducts due to formation of channel inside it and the output of the inverter is then pulled down to logic zero.



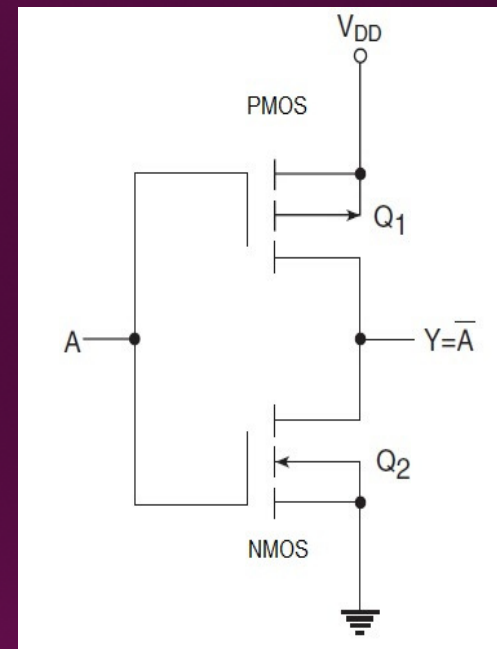
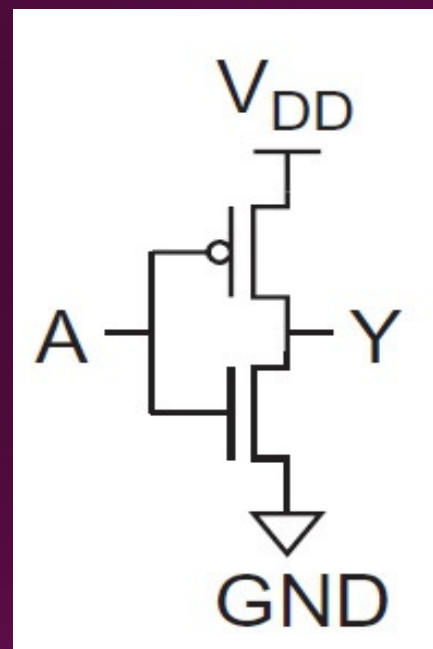
CMOS Inverter

Simplified symbol and switch state has been shown below;



- The bar at the top indicates V_{DD} and the triangle at the bottom indicates GND.
- When the input A is 0, the nMOS transistor is OFF and the pMOS transistor is ON.
- Thus, the output Y is pulled up to 1 because it is connected to V_{DD} but not to GND.
- Conversely, when A is 1, the nMOS is ON, the pMOS is OFF, and Y is pulled down to '0.'

CMOS Inverter

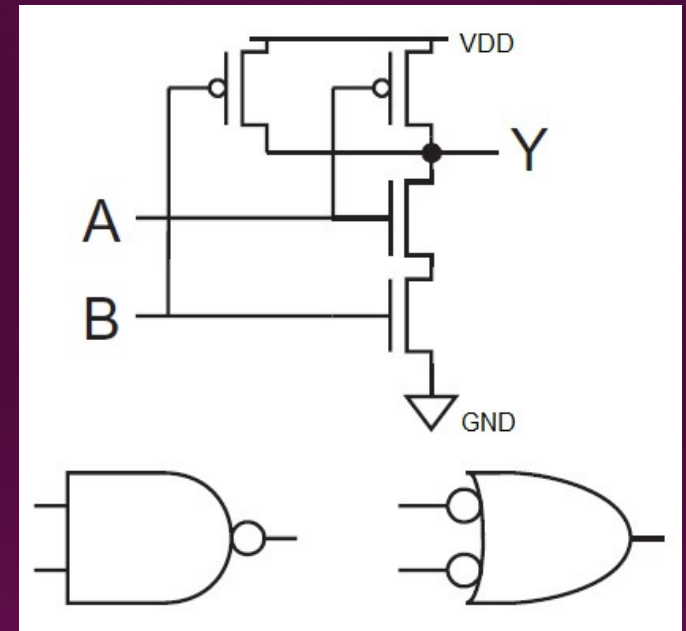


A	Y
0	1
1	0

Truth Table of CMOS NOT gate

CMOS NAND – Logical Circuit

It consists of two series nMOS transistors between Y and GND and two parallel pMOS transistors between Y and VDD.



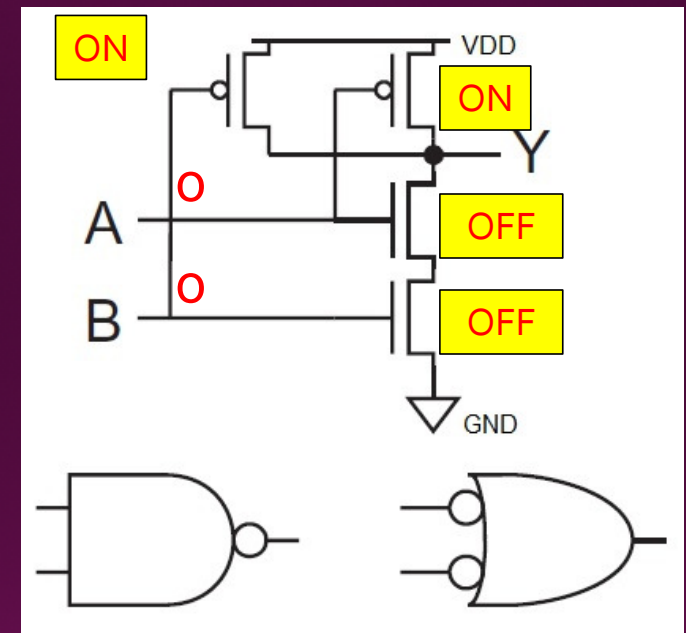
A	B	Pull-Down Network	Pull-Up Network	Y
0	0	OFF	ON	1
0	1	OFF	ON	1
1	0	OFF	ON	1
1	1	ON	OFF	0

Truth Table of CMOS NAND gate

CMOS NAND – working Principle

- When input A or B is 0, nMOS of OFF.

So there is no channel between drain and source and hence pull down network is off.



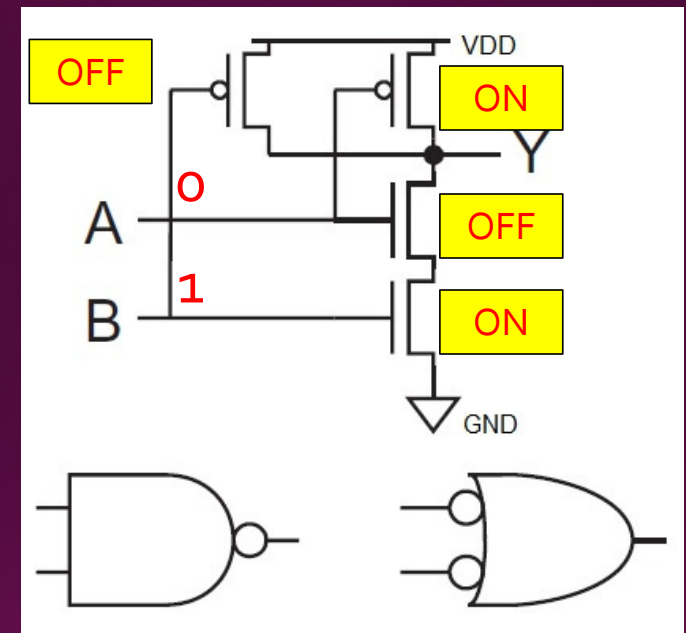
- But the pullup network is ON. Hence the V_{DD} is connected to the output Y.

A	B	Pull-Down Network	Pull-Up Network	Y
0	0	OFF	ON	1
0	1	OFF	ON	1
1	0	OFF	ON	1
1	1	ON	OFF	0

Truth Table of CMOS NAND gate

CMOS NAND – working Principle

- When input A is Low Logic 0 and B is at High logic-1, one of the nMOS is ON and the other is OFF. So, there is no channel between drain and source and hence pull down network is off.
- Due to the input one of the pMOS is ON (With gate input at logic low) and the other is OFF.

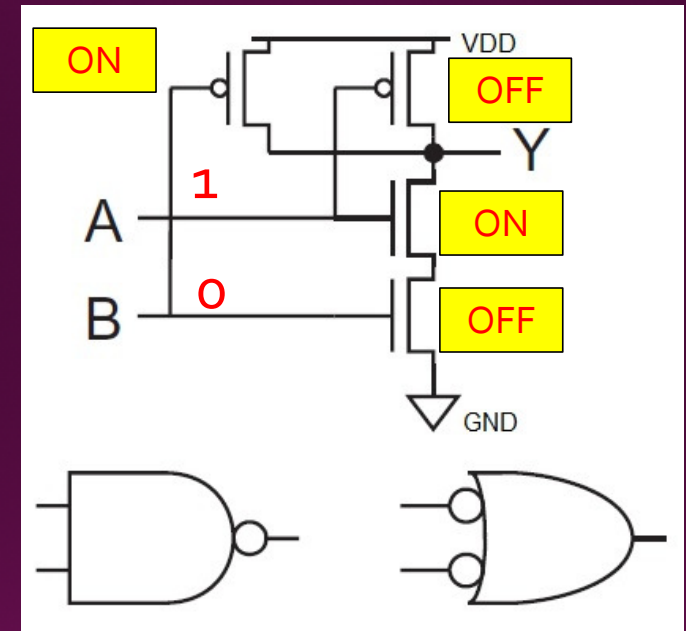


A	B	Pull-Down Network	Pull-Up Network	Y
0	0	OFF	ON	1
0	1	OFF	ON	1
1	0	OFF	ON	1
1	1	ON	OFF	0

Truth Table of CMOS NAND gate

CMOS NAND – working Principle

- When input A is High Logic and B is at low logic-0, one of the nMOS is ON and the other is OFF. So, there is no channel between drain and source and hence pull down network is off.
- Due to the input one of the pMOS is ON (With gate input at logic low) and the other is OFF.

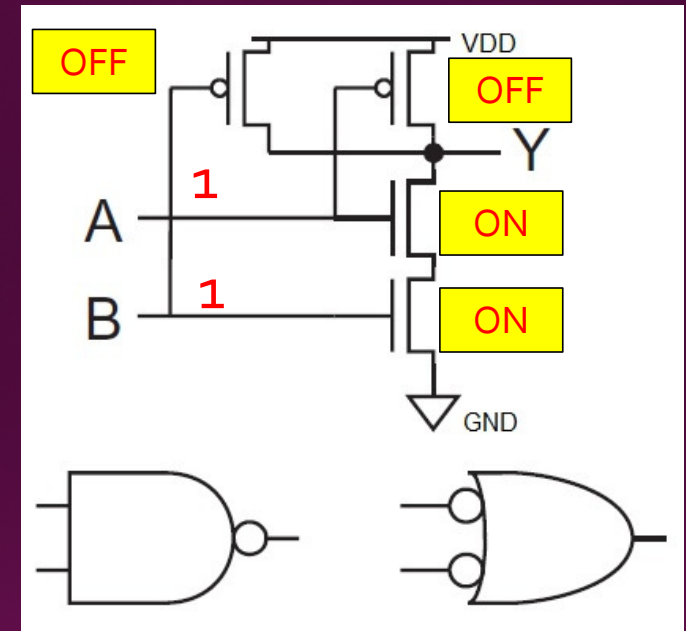


A	B	Pull-Down Network	Pull-Up Network	Y
0	0	OFF	ON	1
0	1	OFF	ON	1
1	0	OFF	ON	1
1	1	ON	OFF	0

Truth Table of CMOS NAND gate

CMOS NAND – working Principle

- When input A & B are at High Logic both the nMOS is ON. So, there is a channel between drain and source and hence pull down network is ON.
- Due to the input both the pMOS is OFF. Output Y is cutoff from the supply voltage V_{DD} .



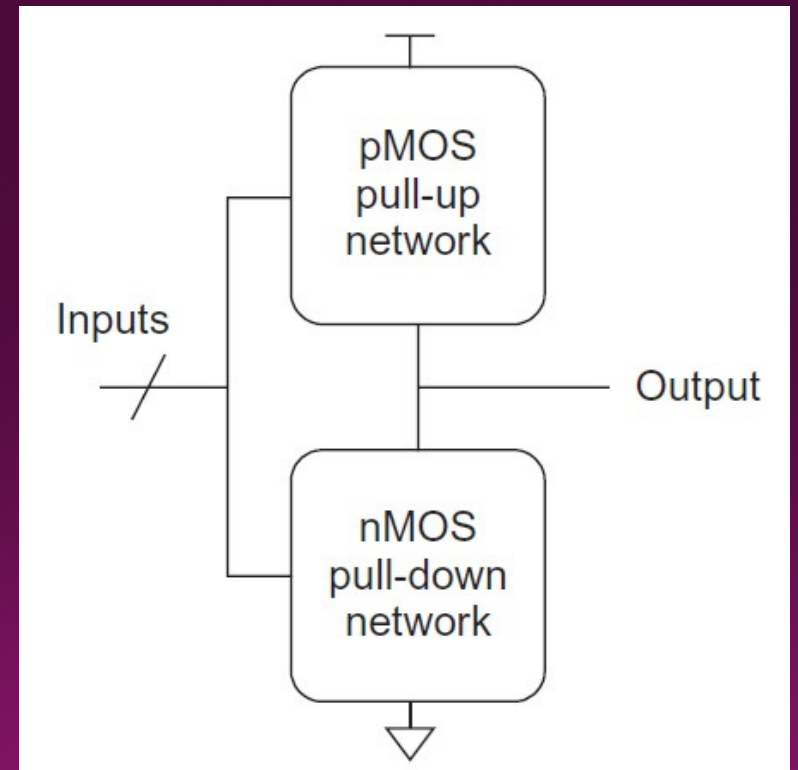
A	B	Pull-Down Network	Pull-Up Network	Y
0	0	OFF	ON	1
0	1	OFF	ON	1
1	0	OFF	ON	1
1	1	ON	OFF	0

Truth Table of CMOS NAND gate

CMOS Push Pull network and combinations

pMOS are always used as pull up network element.

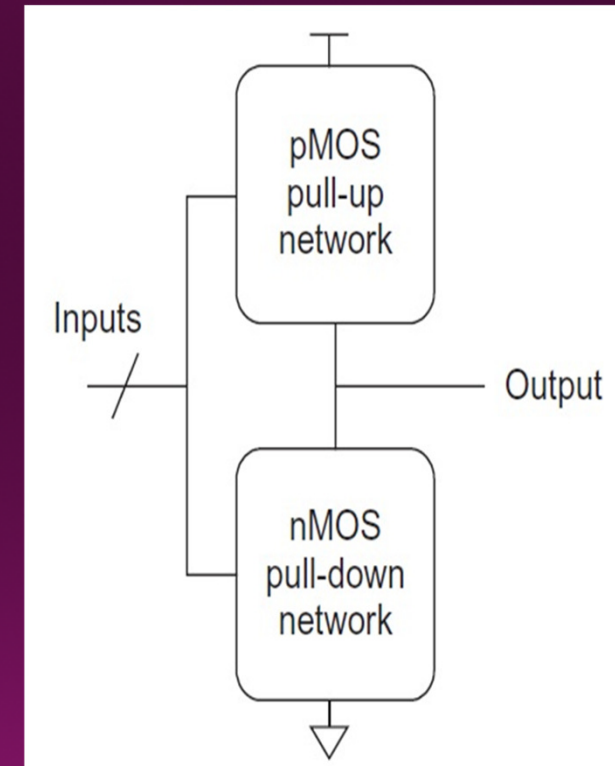
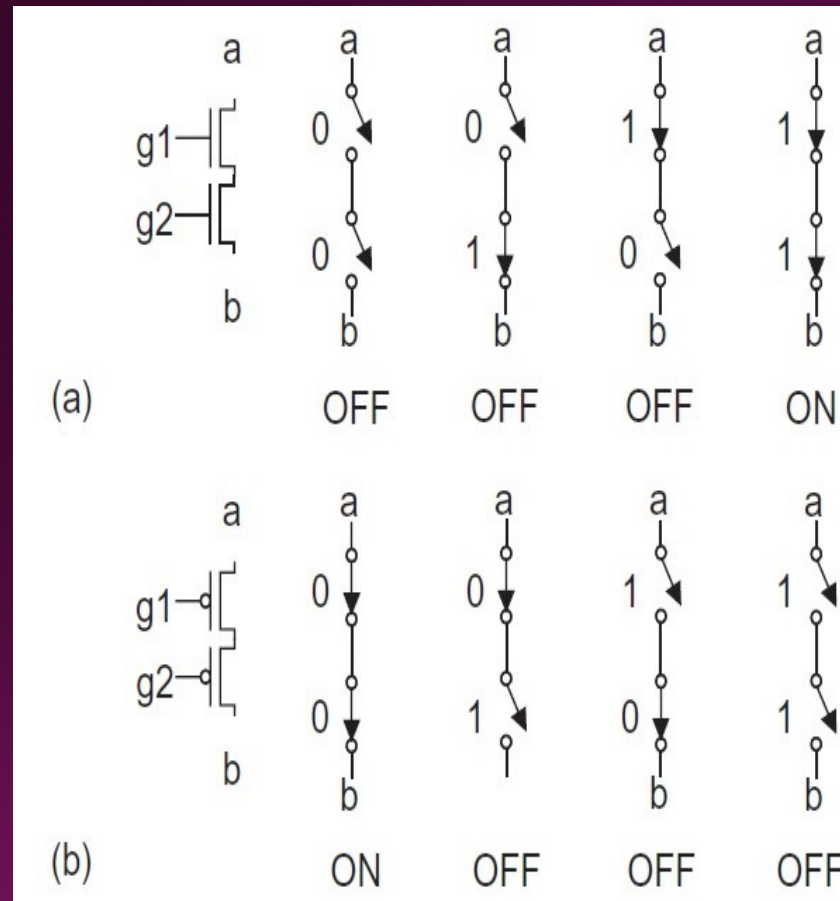
nMOS are in contrast used as pull down network.



CMOS Push Pull network and combinations

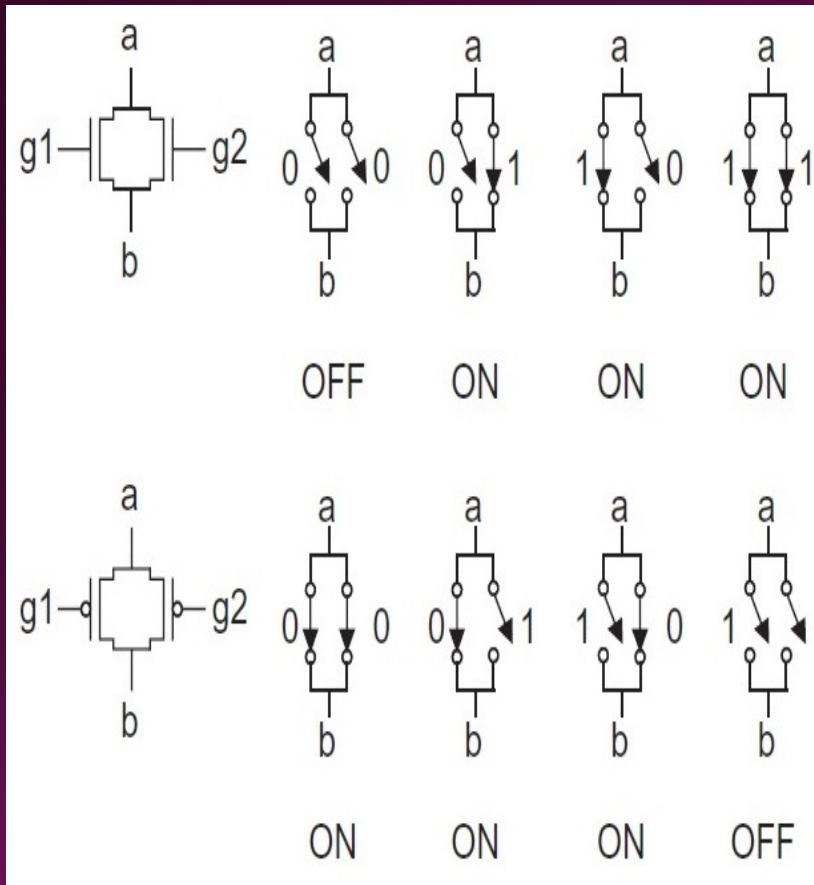
nMOS in series combination

pMOS in series combination

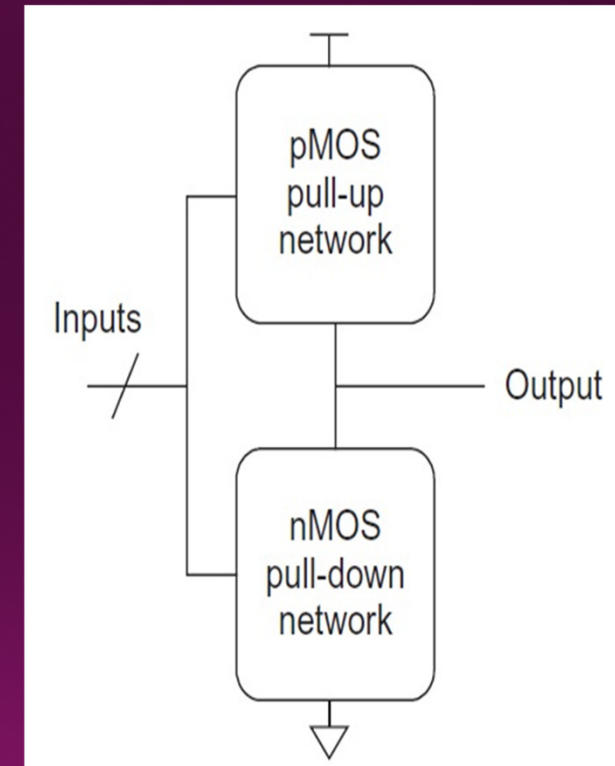


CMOS Push Pull network and combinations

nMOS in parallel combination

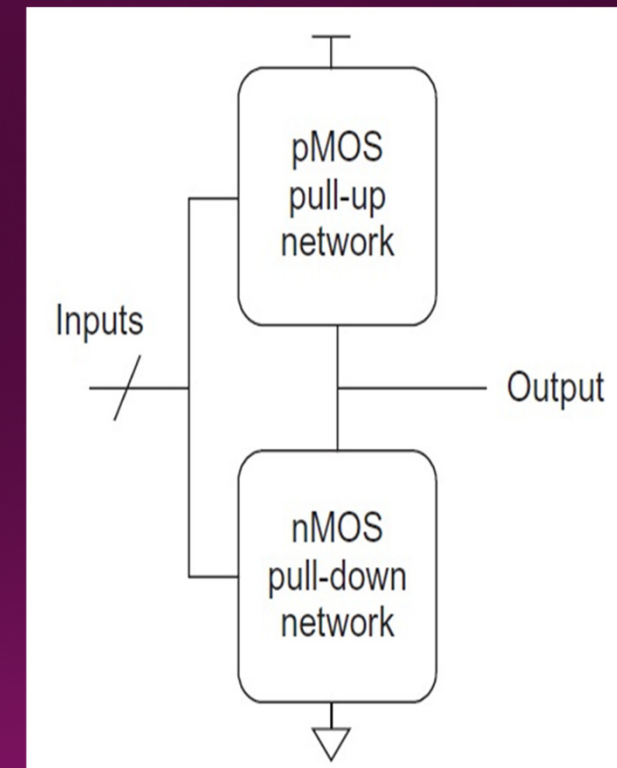


pMOS in parallel combination



CMOS Push Pull network and combinations

	pull-up OFF	pull-up ON
pull-down OFF	Z	1
pull-down ON	0	crowbarred (X)



Compound CMOS Logic gates

A **compound gate** performing a more complex logic function in a single stage of logic is formed by using a combination of series and parallel switch structures.

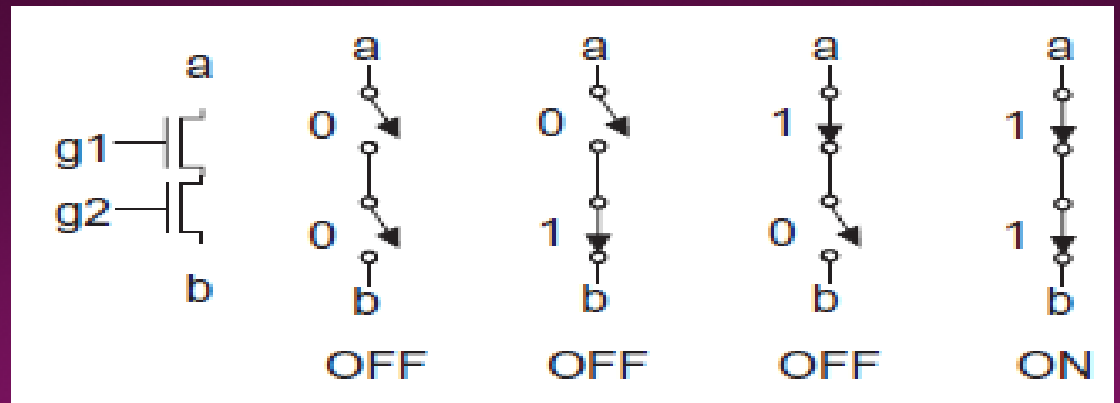
$$Y = \overline{(A \cdot B) + (C \cdot D)}$$

- This function is sometimes called **AND-OR-INVERT-22**, or **AOI22** because it performs the NOR of a pair of 2-input ANDs.
- For the nMOS pull-down network, take the un-inverted expression $((A \cdot B) + (C \cdot D))$ indicating when the output should be pulled to '0.'
- The AND expressions $(A \cdot B)$ and $(C \cdot D)$ may be implemented by series connections of switches.

Compound CMOS Logic gates

$$Y = \overline{(A \cdot B) + (C \cdot D)}$$

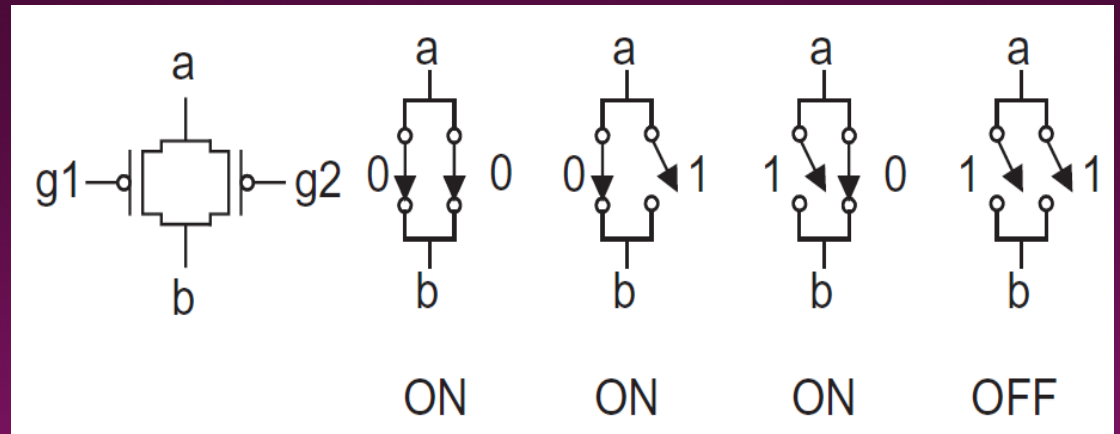
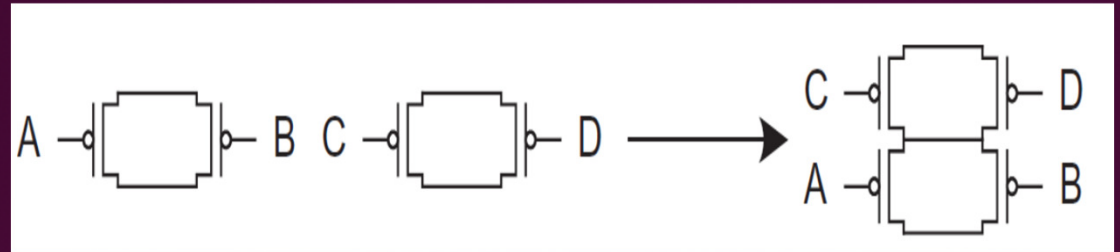
The AND expressions $(A \cdot B)$ and $(C \cdot D)$ may be implemented by series connections of switches.



Compound CMOS Logic gates

$$Y = \overline{(A \cdot B) + (C \cdot D)}$$

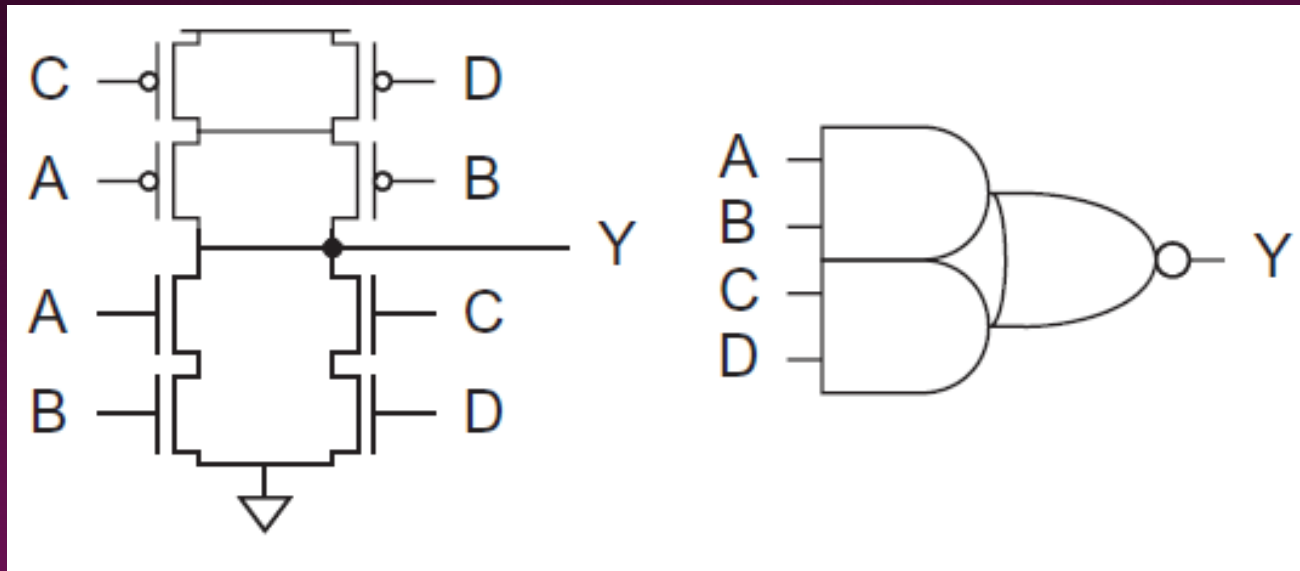
Now ORing the result requires the parallel connection of these two structures.



Compound CMOS Logic gates

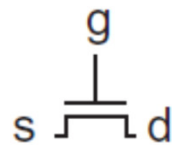
$$Y = \overline{(A \cdot B) + (C \cdot D)}$$

AND-OR-INVERT-22, or AOI22

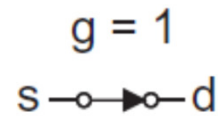
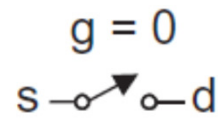


Pass Transistors and Transmission Gates

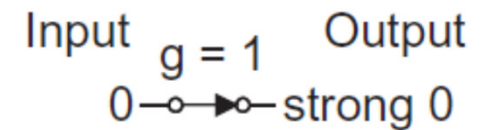
nMOS



(a)



(b)

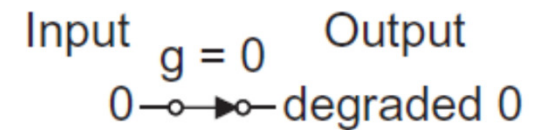
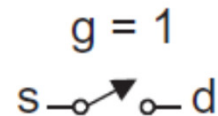
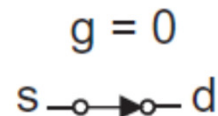
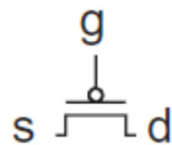


0 → strong 0

g = 1
1 → degraded 1

(c)

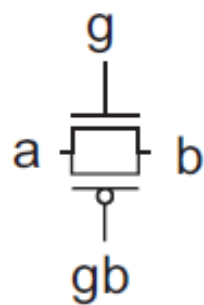
pMOS



0 → degraded 0

g = 0
1 → strong 1

Pass Transistors and Transmission Gates



(a)

$g = 0, gb = 1$
 $a \rightarrow b$

$g = 1, gb = 0$
 $a \rightarrow b$

(b)

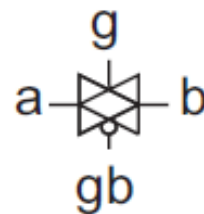
Input

Output

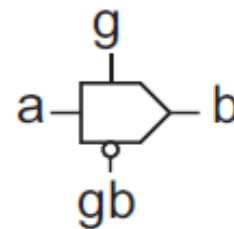
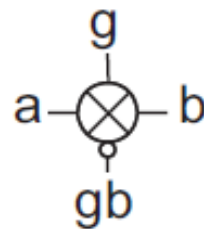
$g = 1, gb = 0$
 $0 \rightarrow \text{strong } 0$

$g = 1, gb = 0$
 $1 \rightarrow \text{strong } 1$

(c)

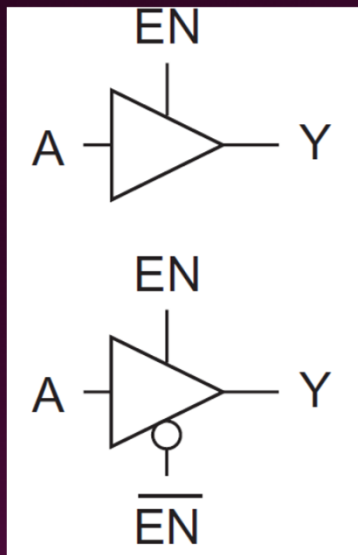


(d)



Transmission gate

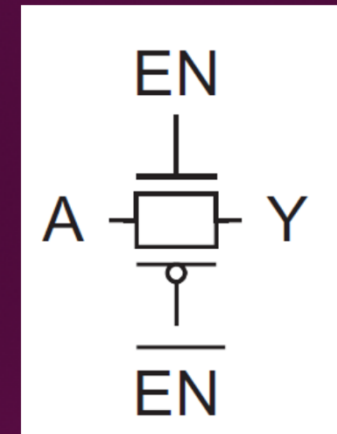
Tristate Buffer



$EN / \overline{EN}$	A	Y
0 / 1	0	Z
0 / 1	1	Z
1 / 0	0	0
1 / 0	1	1

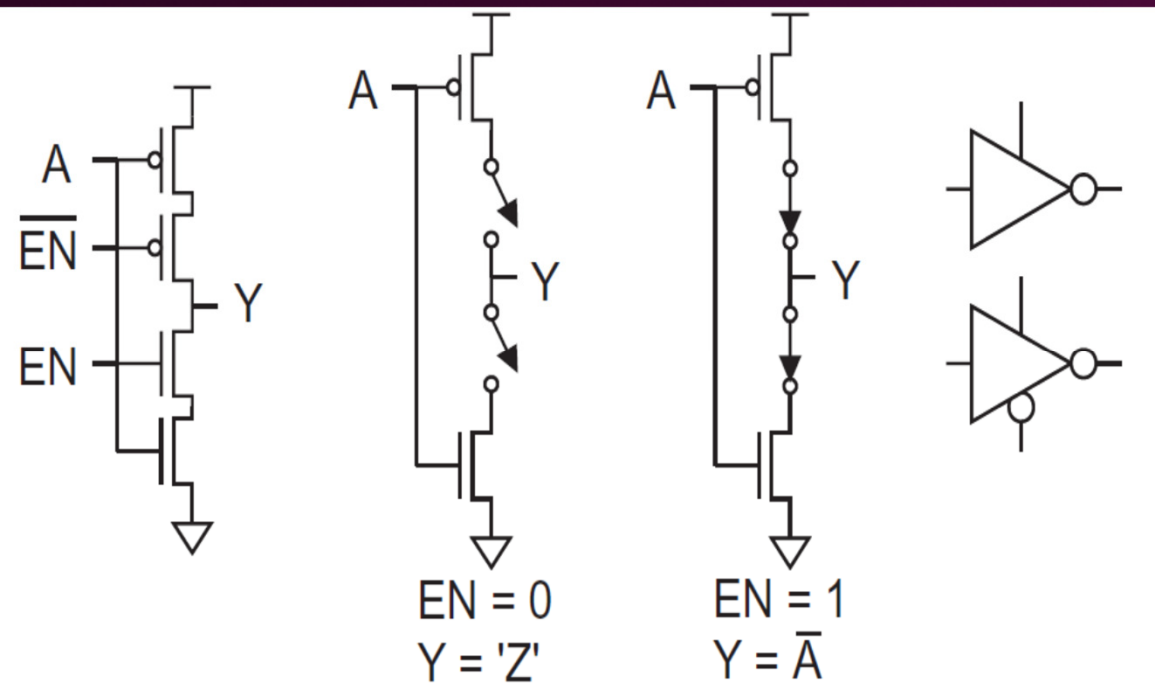
Tristate Buffer using Transmission Gate

- The transmission gate in the figure on the right has the same truth table as a tristate buffer.
- It only requires two transistors/MOSFETS but it is a non-restoring circuit.
- If the input is noisy or otherwise degraded, the output will receive the same noise.



EN / $\overline{\text{EN}}$	A	Y
0 / 1	0	Z
0 / 1	1	Z
1 / 0	0	0
1 / 0	1	1

Tristate Buffer



EN	$\overline{EN}$	$\overline{A}$	Y
0	1	0	Z
0	1	1	Z
1	0	0	1
1	0	1	0

Truth Table of Tri-State Buffer with output inverted.

CMOS problems/task

1. Construct a 2 input NOR CMOS logic circuit. Write the truth table of it.
2. Construct a 3-input NAND CMOS logic circuit. Write down the truth table of it.

Thank You